BIRZEIT UNIVERSITY

Faculty of Engineering & Technology | COMP 433 Software Engineering
Second Semester 2025/2026 | Section 2

Phase 3: Requirements Analysis & Modelling

Business: Cooker Shop | Group 7

#	Name	Student ID	Role in Phase 3
1	Asmaa Jaraba	1231081	Lead: Task 3.2 + Finalization
2	Misk Mashahreh	1231150	Lead: Task 3.4 + Finalization
3	Masa Awad	1220290	Task 3.3 + Document Assembly
4	Anwar Atawna	1222275	Task 3.3 + Assembly Support
5	Abdel Rahman Sabbagh	1203015	Lead: Task 3.1 + Task 3.3

Instructor: Dr. Adel Taweel

Section : 2

Submitted by: Group 7 Developer Group — Cooker Shop Project

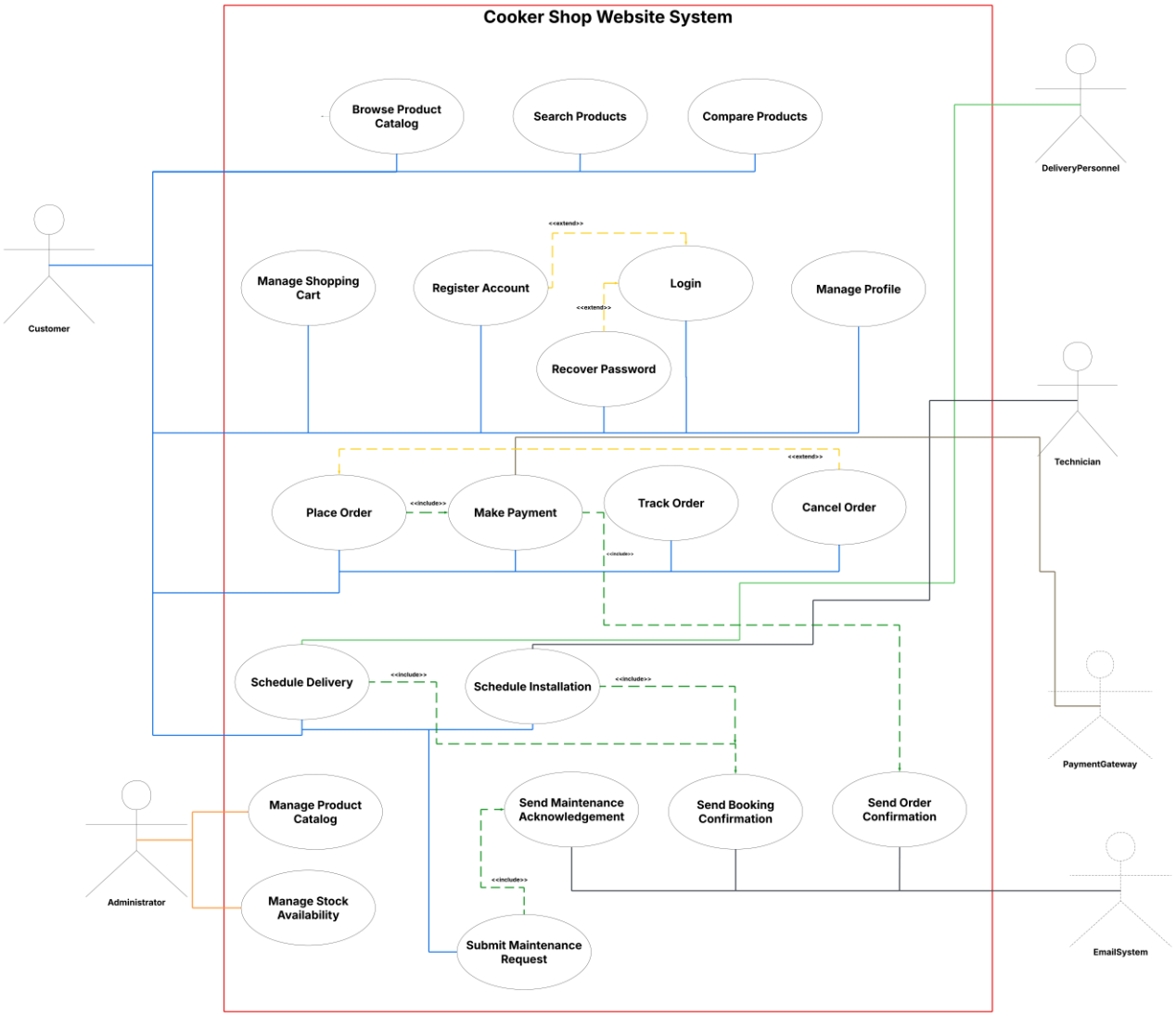
Task 3.1 — Actor Analysis

Abdel Rahman Sabbagh (1203015)	LEAD — Task 3.1 Actor Analysis

Actor Name	Type	Semantic Description
Customer	Principal	This actor represents any individual, household, restaurant, or business that uses the Cooker Shop website to browse products, search and compare cookers, manage their shopping cart, register or log in, place orders, make payments, schedule delivery and installation, and submit maintenance requests.
Administrator	Principal	This actor represents the store manager who uses the admin dashboard to manage the product catalog, add, update, or remove cooker products, control stock availability, and process customer orders.
DeliveryPersonnel	Principal	This actor represents the delivery workers who collect scheduled delivery bookings from the system and execute deliveries to customers at their specified addresses.
Technician	Principal	This actor represents the installation engineers who collect scheduled installation appointments from the system and perform cooker installation services at customer premises.
PaymentGateway	Secondary	This actor represents the external secure payment processing system that handles credit and debit card transactions on behalf of the Cooker Shop website.
EmailSystem	Secondary	This actor represents the external email service that dispatches automated notifications such as order confirmations, booking confirmations, and maintenance request acknowledgements to customers.

Task 3.2 — Use Case Diagram [Group Task — Lead: Asmaa Jaraba]

Asmaa Jaraba (1231081)	LEAD — Task 3.2 Use Case Diagram



Task 3.3 — Use Case Specifications [Individual Task]

3.3.1 Asmaa Jaraba — Place Order

Asmaa Jaraba (1231081)	Individual — Place Order

Field	Content
System Name	Cooker Shop Website System
Use Case Title	Place Order
Actors	Customer, PaymentGateway, EmailSystem
Data	Order details, delivery address, phone number, selected products, quantities, payment details
Stimulus / Trigger	Customer clicks the 'Confirm Order' button after reviewing the order summary
Pre-conditions	1. Customer is logged in to a registered account. 2. Shopping cart contains at least one in-stock product. 3. System is online and accessible.
Description	This use case describes the process by which a registered customer confirms and submits an order for one or more cooker products. The system validates the order details, processes the payment through the Payment Gateway, generates a unique order identifier, and dispatches a confirmation email to the customer via the EmailSystem.
Workflow / Flow of Events	Normal Flow: 1. The customer reviews the order summary showing product details, quantities, total cost, and delivery address. 2. The customer clicks "Confirm Order". 3. The system validates all required fields including delivery address and phone number. 4. The system checks that all products in the cart are in stock. 5. The system passes the order to the PaymentGateway for payment processing. 6. The PaymentGateway confirms successful payment. 7. The system generates a unique order ID and records the order. 8. The EmailSystem dispatches an order confirmation to the customer's registered email address. 9. The system displays a success message with the order ID to the customer.

	Error Flow: 3a. If any required field is missing or invalid → the system highlights the missing field and displays an error message. The use case returns to step 1. 4a. If one or more products are out of stock → the system rejects the order, notifies the customer, and returns them to the cart. 6a. If payment fails → the PaymentGateway returns a failure response → the system notifies the customer and returns to step 1. The order is not recorded.
Post-conditions	1. A new order is recorded in the system with a unique order ID. 2. The order status is set to "Placed". 3. An order confirmation email has been sent to the customer. 4. Stock availability is updated for all ordered products.
Comments	Customer must be authenticated before placing an order. Out-of-stock products cannot be ordered.

3.3.2 Misk Mashahreh — Make Payment (UC8)

Misk Mashahreh (1231150)	Individual — Make Payment (UC8)

Field	Content
System Name	Cooker Shop Website System
Use Case Title	Make Payment
Actors	Customer, Payment Gateway
Data	Card information, Order information, Customer Information, Payment Transaction information.
Stimulus / Trigger	Customer submits payment information.
Pre-conditions	1. Customer is logged in. 2. The customer has a pending order with at least one item. 3. The final order total has been calculated, displayed, and locked.
Description	The customer pays for an order through an external payment gateway. The system validates the payment transaction, records the payment details, and updates the order status accordingly.
Workflow / Flow of Events	**Normal Flow:** 1. Customer places an order. 2. The system checks if the order is valid. 3. The system displays available payment options 4. Customer selects a payment method and enters payment information. 5. The system checks whether the payment gateway is available. 6. The system sends a payment request to the Payment Gateway. 7. The Payment Gateway validates the payment information. 8. The Payment Gateway returns an approval response 9. The system records the payment transaction successfully. 10. The system updates the order status to “Paid”. 11. The system sends an order confirmation notification to the customer. **Alternative / Error Flow:** **A2:** if the order is not valid, the system displays an “Invalid Order” message and terminates the process. A5:If the payment gateway is not available, the system displays a “Payment Service Unavailable” message and terminates the process. A7: If the payment is not approved, the Payment Gateway returns a rejection response and the system displays a “Payment Failed” message.

Post-conditions	1. Payment transaction is recorded successfully. 2. Order status is updated to “Paid”. 3. Order confirmation is sent to the customer. 4. Customer is able to track the order.
Comments	Payment is processed through a secure external payment gateway. Full card details are not permanently stored by the system for security purposes.

3.3.3 Masa Awad — Schedule Delivery & Installation

Masa Awad (1220290)	Individual — Schedule Delivery & Installation

Field	
System Name	Cooker Shop Website System
Use Case Title	Schedule Delivery (UC14)
Actors	Customer, DeliveryPersonnel, EmailSystem
Data	Available delivery time slots, customer delivery address, preferred delivery date and time
Stimulus / Trigger	Customer proceeds to schedule delivery after successful payment
Pre-conditions	1. Order has been successfully placed and payment confirmed (UC9 and UC10 completed). 2. At least one delivery time slot is available for the customer's location.
Description	This use case describes the process by which a customer schedules the delivery and installation of a purchased cooker after successful payment. The customer selects a preferred delivery date and time slot from the available options. The system validates availability, assigns suitable DeliveryPersonnel, records the booking, and sends a confirmation notification to the customer. The system ensures that delivery personnel are not double-booked for the same time slot.
Workflow / Flow of Events	**Normal Flow:** 1. The customer accesses the delivery scheduling page after successful payment. 2. The system retrieves and displays all available delivery and installation time slots for the customer's location. 3. The customer selects a preferred delivery date and time slot. 4. The system verifies that the selected slot is still available. 5. The system assigns suitable DeliveryPersonnel for the selected slot. 6. The system records the delivery and installation booking. 7. The EmailSystem sends a confirmation email containing the delivery date, time, and booking details. 8. The system displays a confirmation message to the customer. 9. The use case ends successfully. **Alternative / Error Flow:**

	4a. If the selected delivery slot is no longer available, the system informs the customer and displays alternative available slots. The use case returns to step 3. 5a. If no DeliveryPersonnel are available for the selected slot, the system notifies the customer and requests selection of another date or time. The use case returns to step 3. 7a. If the EmailSystem fails to send the confirmation email, the booking remains recorded successfully and the system displays the booking confirmation on screen while logging the email failure for later retry.
Post-conditions	1. A delivery and installation booking has been successfully recorded in the system. 2. The selected delivery time slot is marked as reserved. 3. Assigned DeliveryPersonnel are linked to the booking. 4. A delivery confirmation notification has been generated and sent to the customer whenever possible. 5. The order status is updated to “Delivery Scheduled”.
Comments	Customers may book installation simultaneously with or independently from delivery. Double-booking of the same personnel is not permitted.

3.3.4 Anwar Atawna — Submit Maintenance Request

Anwar Atawna (1222275)	Individual — Submit Maintenance Request

Field	
System Name	Cooker Shop Website System
Use Case Title	Submit Maintenance Request (UC19)
Actors	Customer, EmailSystem
Data	Invoice number, product model number, written description of the issue, photo or video attachment
Stimulus / Trigger	Customer navigates to the maintenance request form on the website
Pre-conditions	1. Customer is logged in to a registered account. 2. Customer has a valid invoice number from a previous purchase recorded in the system.
Description	This use case allows a registered customer to submit a maintenance request for a cooker product purchased from the website. The customer enters the invoice number, product model number, and a clear description of the issue. The customer may also upload a photo or video to help explain the problem. The system validates the entered information, checks that the invoice exists, and confirms that the product is still within the warranty period. If everything is valid, the system creates a maintenance request and generates a unique maintenance ticket ID for follow-up.
Workflow / Flow of Events	**Normal Flow:** 1. The customer logs in to the website using a registered account. 2. The customer navigates to the maintenance request form. 3. The customer enters the invoice number and product model number. 4. The customer writes a clear description of the product issue. 5. The customer optionally uploads a photo or video showing the issue. 6. The customer clicks the “Submit Request” button. 7. The system validates the entered information and verifies that the invoice number exists in the system. 8. The system checks that the product is still within the warranty period. 9. The system creates a maintenance request, generates a unique ticket ID, sends a confirmation email, and displays a success message to the customer. **Alternative / Error Flow:** 3a. If the invoice number or product model number is missing or invalid, the system displays an error message and asks the customer to enter valid information. 4a. If the issue description is empty or unclear, the system asks the customer to provide a clearer description of the problem. 5a. If the uploaded photo or video format is not supported, the system displays an error message and asks the customer to upload a valid file.

	8a. If the product is outside the warranty period, the system rejects the request and informs the customer that maintenance requests are accepted only for products within the warranty period..
Post-conditions	**Post-conditions:** 1. A new maintenance request is stored in the system. 2. A unique maintenance ticket ID is generated for the request. 3. The request status is set to “Pending Review”. 4. The customer receives a confirmation email containing the maintenance ticket ID. 5. The maintenance team can view the submitted request and follow up with the customer.
Comments	Requests accepted only for products within their warranty period. A unique maintenance ticket ID is generated for each accepted request.

3.3.5 Abdel Rahman Sabbagh — Manage Product Catalog

Abdel Rahman Sabbagh (1203015)	Individual — Manage Product Catalog

Field	
System Name	Cooker Shop Website System
Use Case Title	Manage Product Catalog (UC17)
Actors	Administrator
Data	Product name, brand, cooker type, size, price, features, technical specifications, images, and availability status.
Stimulus / Trigger	The administrator needs to add a new cooker product, update an existing product, deactivate or remove a product, or change the product stock availability.
Pre-conditions	1. The administrator is authenticated. 2. The administrator has valid admin privileges. 3. The system is online and the product management service is available.
Description	This use case describes how the administrator manages cooker products in the Cooker Shop Website System. The administrator can add a new product, update existing product information, deactivate or remove a product, and update stock availability. The system validates the entered product data before saving changes. After a successful update, the product catalog and product availability are updated for customers.
Workflow / Flow of Events	**Normal Flow:** 1. The administrator starts the product management function. 2. The system displays the available product management options: add product, update product, deactivate product, remove product, or update stock status. 3. The administrator selects the required operation. 4. If adding a new product, the administrator enters the product name, brand, cooker type, size, price, features, specifications, images, and availability status. 5. If updating an existing product, the administrator selects the product and modifies the required details. 6. If changing stock status, the administrator selects the product and marks it as In Stock or Out of Stock. 7. The system validates all required product data. 8. The system checks that the product identifier or model is not duplicated when adding a new product. 9. The system saves the product changes. 10. The system updates the product catalog and stock availability. 11. The system records the action with the administrator identity and update time. **Alternative / Error Flow:** 7a. If required product data is missing or invalid, the system rejects the update and asks the administrator to correct the missing or invalid fields. 8a. If a duplicate product identifier or model is entered, the system rejects the new product and asks for a unique product identifier. 9a. If the administrator tries to permanently delete a product linked to an active or previous order, the system prevents permanent deletion and allows deactivation instead. 9b. If an uploaded image file is invalid, unsupported, or cannot be uploaded, the system rejects the file and asks the administrator to upload a valid image.

Post-conditions	1. A new product is added, an existing product is updated, or a product is deactivated/removed according to the administrator's action. 2. The product catalog reflects the latest valid product information. 3. Product availability status is updated. 4. Products marked as Out of Stock cannot be purchased by customers. 5. The system stores a record of the administrator's action.
Comments	Only authenticated administrators with proper privileges can manage the product catalog. Product deletion should not affect confirmed or historical orders. Deactivation is preferred when a product has existing order records.

Task 3.4 — Activity Diagram with Swimlanes [Group Task — Lead: Misk Mashahreh]

Misk Mashahreh (1231150)	LEAD — Task 3.4 Activity Diagram

